Friedman and colleagues were studying ultraviolet radiation not only from the Sun, but also from stars. X-rays, however, were another matter. Each second the Sun dumped 1 million X-rays onto a square centimeter of their Geiger counter, so detecting the Sun with X-rays was relatively easy. Theoretical estimates, however, suggested that the brightest X-ray stars would be 1 billion times fainter than the Sun. To detect so faint a star would require an X-ray detector 10 million times more sensitive than those that Friedman was flying in 1958. Such an improvement was a tall order, but not impossible.

By 1962, the detectors had been improved 10,OOO-fold. With just another factor of a thousand to go, other research groups, impressed by Friedman's progress, were beginning to compete with him. One, a team led by Riccardo Giacconi, would become a formidable competitor.

In a peculiar way, Zel'dovich may have shared responsibility for Giacconi's success. In 1961, the Soviet Union unexpectedly abrogated a mutual Soviet/American three-year moratorium on the testing of nuclear weapons, and tested the most powerful bomb ever exploded by humans—a bomb designed by Zel'dovich's and Sakharov's teams at the Installation (Chapter 6). In panic, the Americans prepared new bomb tests of their own. These would be the first American tests in the era of Earth-orbiting spacecraft. For the first time it would be possible to measure, from space, the X-rays, gamma rays, and high-energy particles emerging from nuclear explosions. Such measurements would be crucial for monitoring future Soviet bomb tests. To make such measurements on the impending series of American tests, however, would require a crash program. The task of organizing and leading it went to Giacconi, a twenty-eight-year-old experimental physicist at American Science and Engineering (a private Cambridge, Massachusetts, company), who had recently begun to design and fly X-ray detectors like Friedman's. The U.S. Air Force gave Giacconi all the money he needed, but little time. In less than a year, he augmented his six-person X-ray astronomy team by seventy new people, designed, built, and tested a variety of weapons-blast monitoring instruments, and flew them with a 95 percent success rate in twenty-four rockets and six satellites. This experience molded the core members of his group into a loyal, dedicated, and highly skilled team, ideally primed to beat all competitors in the creation of X-ray astronomy.